

TeamFlow – AI-Powered Team Collaboration & Project Monitoring Platform

Introduction

TeamFlow is a web-based application that combines **project management, team collaboration, task tracking, AI assistance, and email notifications** into one platform. It is designed for organizations where project managers need better visibility into project progress while enabling employees to collaborate efficiently.

Unlike a normal chat application, TeamFlow integrates project management with communication. Every project has dedicated workspaces where tasks, files, discussions, and updates are managed in one place.

Problem Statement

In many organizations:

- Project updates are scattered across different applications.
- Managers struggle to track employee progress.
- Important discussions get lost in chats.
- Deadlines are missed due to lack of reminders.
- Preparing project status reports is time-consuming.

A centralized platform is needed that combines project management, communication, AI-generated summaries, and automated notifications.

Objective

The objective of TeamFlow is to:

- Manage projects centrally.
- Improve collaboration between employees.
- Track task progress in real time.
- Help managers monitor project status.
- Generate AI-based summaries.
- Send automatic email notifications.
- Improve productivity and reduce project delays.

Target Users

Admin

Responsible for managing the organization.

Functions:

- Create users
- Assign roles
- Create projects
- Assign Project Managers
- View organization analytics
- Monitor all projects

Project Manager

Responsible for managing project execution.

Functions:

- Create tasks
- Assign employees
- Monitor project discussions
- Track employee performance
- Generate project reports
- View AI-generated summaries

Employee

Responsible for completing assigned work.

Functions:

- View assigned tasks
- Update task progress
- Participate in project chats
- Upload project files
- Submit completed work

Core Modules

Module 1 – Authentication

Features

- User Login
- Secure Registration (Admin creates users)
- JWT Authentication
- Forgot Password
- Change Password

Module 2 – User Management

Admin can

- Add Employees
- Edit Employee Details
- Delete Employees
- Assign Roles
- Activate/Deactivate Users

Module 3 – Project Management

Manager can

- Create Project
- Edit Project
- Archive Project
- Assign Members
- Set Start Date
- Set End Date
- Update Project Status

Project Status

- Planning
- In Progress
- Testing
- Completed
- Delayed

Module 4 – Task Management

Manager can

- Create Tasks
- Assign Employees
- Set Due Dates
- Set Priority
- Attach Files
- Track Progress

Employee can

- Accept Task
- Update Progress
- Submit Work
- Add Comments

Priority

- Low

- Medium
- High
- Critical

Module 5 – Team Collaboration

Each project has its own workspace.

Features

- Project Group Chat
- One-to-One Chat (optional)
- File Sharing
- Image Sharing
- Document Sharing
- Task Discussions

Managers can access official project conversations to support project execution. This monitoring is intended for project transparency within the workspace rather than hidden personal surveillance.

Module 6 – Dashboard

Admin Dashboard

Displays

- Total Employees
- Total Managers
- Active Projects
- Completed Projects
- Delayed Projects
- Overall Performance
- Monthly Statistics

Manager Dashboard

Displays

- Assigned Projects
- Pending Tasks
- Team Performance
- Employee Activity
- AI Summary
- Project Progress

Employee Dashboard

Displays

- My Tasks
- Due Today
- Upcoming Deadlines
- Notifications
- Recent Chats

Module 7 – Reports

Generate

- Project Report
- Employee Report
- Task Completion Report
- Performance Report

Export

- PDF
- Excel

Module 8 – Notifications

Automatic notifications for

- New Task Assigned
- Deadline Reminder
- Project Updates
- Task Completed
- Manager Comments

Module 9 – File Management

Upload

- PDF
- Word
- Excel
- PPT
- Images
- ZIP Files

Download

- Reports
- Project Files
- Shared Documents

API Integration

API 1 – OpenAI API

Purpose

- Summarize long project chats.
- Generate daily progress reports.
- Provide weekly project summaries.
- Suggest task priorities.
- Answer project-related questions using project context.

Example

Manager clicks **Generate Summary**

Output

Today's Progress

- Login Module Completed
- Database Schema Updated
- Payment Module Pending
- Two deadlines approaching this week

API 2 – Email API (EmailJS / SendGrid / Brevo)

Purpose

Automatically sends emails when:

- A new task is assigned.
- A project deadline is approaching.
- A task is completed.
- Password reset is requested.
- Important announcements are made.

Example

Employee receives:

Subject: New Task Assigned

Hello John,

You have been assigned the Authentication Module.

Due Date: 10 July 2026

Database Design

Users

- User ID

- Name
- Email
- Password
- Role
- Department

Projects

- Project ID
- Project Name
- Description
- Start Date
- End Date
- Status

Project Members

- Member ID
- Project ID
- User ID

Tasks

- Task ID
- Project ID
- Assigned To
- Title
- Description
- Priority
- Status
- Deadline
- Progress

Chat Messages

- Message ID
- Sender ID
- Project ID
- Message
- Timestamp

Files

- File ID
- Project ID
- Uploaded By
- File Name
- File URL

Notifications

- Notification ID
- User ID
- Message
- Status

Reports

- Report ID
- Project ID
- Generated By
- Generated Date

Technology Stack

Frontend

- ReactJS
- HTML
- CSS
- JavaScript
- Bootstrap

Backend

- Spring Boot
- Spring Security
- Spring Data JPA
- REST APIs

Database

- PostgreSQL

Authentication

- JWT Authentication

Real-Time Communication

- WebSocket (STOMP)

APIs

- OpenAI API
- Email API (EmailJS / SendGrid / Brevo)

System Workflow

1. Admin logs in.
2. Admin creates users and assigns roles.
3. Admin creates a project and assigns a Project Manager.
4. The Project Manager creates tasks and assigns them to employees.
5. Employees receive email notifications about assigned tasks.
6. Employees update task progress, upload files, and communicate in the project workspace.
7. The Project Manager monitors progress, discussions, and pending work through the dashboard.
8. The OpenAI API generates chat summaries and highlights important updates.
9. Reports are generated based on project and employee performance.
10. When tasks are assigned, deadlines approach, or projects are updated, the Email API automatically sends notifications.
11. Once all tasks are completed, the Project Manager closes the project, and the Admin can view the final project report.

Future Enhancements

- Video conferencing integration
- Calendar synchronization
- Attendance tracking
- Time tracking (timesheets)
- Kanban board
- Gantt chart
- AI-based project risk prediction
- Mobile application
- Multi-organization support

Expected Outcome

The system will provide a centralized platform where project planning, task management, communication, AI-powered insights, and automated notifications work together. It reduces

manual reporting, improves transparency, keeps team members informed through email alerts, and helps managers make better decisions using AI-generated summaries and analytics.

Project Architecture



